RYERSON UNIVERSITY
DEPARTMENT OF MATHEMATICS, PHYSICS & COMPUTER SCIENCE

PCS125 MIDTERM TEST 17 October 2003

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Student Number	Section	Print First Name	Print Last Name

Please circle your professor's name : M. Buckby J. Hu S. Pejovic G. Teneva J. Brito M. Kolios R. Szmidt

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INSTRUCTIONS:

- This is a multiple-choice test; please select the option which is closest to the correct answer. Circle correct answers in this booklet.
- Transcribe your answers onto the bubble sheet; only the bubble sheet will be marked.
- Do not fold, spindle or mutilate your bubble sheet.
- The formula sheet at the end of the test may be detached and retained. The remaining test pages must not be unstapled. Turn bubble sheet face down when not working on it.
- Only non-programmable calculators are permitted.
- The test consists of 12 questions and is worth 25% of your course grade.
- All questions are of equal weight. Incorrect answers will receive a mark of zero; there are no further punitive deductions.
- At the end of the test, insert your bubble sheet into the test booklet and hand in both.

Please sign here indicating you have read and understood the above instructions.

Signature: _____
(no signature, no mark!)

DO NOT OPEN THE TEST UNTIL YOU ARE TOLD TO DO SO.

[1] An object undergoes simple harmonic motion (SHM) of amplitude A. During one period (one complete cycle), through what total distance does the object move?

(A) 2 A (B) A^2 (C) 4 A (D) A/2 (E) A

10/12

2. [1] A massive particle on a spring is pulled 3.00 cm downward from equilibrium and released at t = 0. The maximum upward acceleration is 0.300 m/s². Neglecting frictional influences, calculate the period (in seconds) of this simple harmonic oscillator.

(A) 0.50 (B) 1.99 (C) 0.63 (D) 1.59 (E) 0.32

A = 3.00 cm
a_{max} = 0.300 m/s²

T = ?
A = 3.00 cm
a_{max} = 0.300 m/s.

$a_{max} = \omega^2 A$

$\frac{0.300 m/s}{0.03 m} = \omega^2$

$\therefore \omega = \sqrt{10}$

$\therefore T = \frac{2\pi}{\omega}$

$\doteq 1.99 s$

3. [1] After a thrilling plunge from a high bridge, a bungee jumper bounces in simple harmonic motion (SHM) at the end of the vertical bungee cord. Jumper-Joe of mass m oscillates with period T. When Jumper-Jane of mass m′ takes her turn on the same bungee cord, she oscillates with a period T′. Determine Jumper-Jane's mass, m′, as a function of (m, T, T′).

m = 100 T = 4
m′ = ? T′ = 3

(A) $m' = 2\pi\left(\frac{TT'}{m}\right)^{1/2}$ (B) $m' = m\left(\frac{T'}{T}\right)^2$ (C) $m' = 2\pi\left(\frac{m}{TT'}\right)^{1/2}$

(D) $m' = m\left(\frac{T}{T'}\right)^2$ (E) $m' = (mTT')^{1/2}$

$T = 2\pi\sqrt{\frac{m}{k}}$

$T' = \sqrt{\frac{m}{k}}$

$T'Tk = \frac{m'}{k}$

$\frac{T}{m} = \frac{T'}{m'}$

$m' = \frac{T'm}{T}$

$m' = m\left(\frac{T'}{T}\right)$

$m' =$

l = 0.400 m

4. [1] A simple pendulum consists of 40.0 cm of string (of negligible mass) and a small 50.0 g ball (of negligible size). When the ball passes through its lowest possible position (equilibrium position), its speed is 60.0 cm/s. Determine the pendulum's maximum angular displacement with respect to the vertical. Note: acceleration due to gravity is $g = 9.80 m/s^2$.

(A) 0.304° (B) 28.6° (C) 17.4° (D) 67.4° (E) 22.6°

$\therefore \cos\theta = \frac{0.41837 m}{0.400 m}$

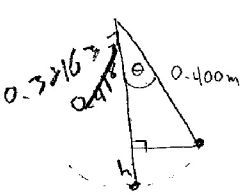
$\doteq 17.4°$

l = 0.400 m
V_{max} = 60.0 cm/s
m = 50.0 g
$T = 2\pi\sqrt{\frac{L}{g}}$
T = 1.27 s.

$gh = \frac{1}{2}v^2$

$h = \frac{\frac{1}{2}v^2}{g}$

$h = 0.01837 m$



$\omega = \frac{2\pi}{T}$

$\omega = 4.9474 rad/s$

$mgh = \frac{1}{2}mv^2$

5. [1] A sinusoidal (harmonic) wave on a rope is described by the wave equation for transverse (vertical) displacement, y:

$$y = 0.05\sin(\overset{k}{9.0}x - \overset{\omega}{3.14}t)$$

where all quantities are in S. I. units. If the mass per unit length is 0.06 kg/m, determine the average power (in Watts) carried by the wave.

(A) 6.1×10^{-3} (B) 5.2×10^{-4} (C) 8.2×10^{-5} (D) 5.2×10^{-3} (E) 2.6×10^{-4}

A = 0.05 m

μ = 0.06 kg/m

k = 9.0

ω = 3.14

$T = \frac{2\pi}{\omega}$

= 2.00 s

$\lambda = \frac{2\pi}{k}$

λ = 0.698 m

∴ v = fλ

$v = \frac{\lambda}{T}$ ∴ $P = \frac{1}{2}\mu\omega^2 A^2 v$

$= \frac{0.698\,m}{2.00\,s}$ = 2.6×10^{-4}

v = 0.349 m/s.

6. [1] A sinusoidal (harmonic) wave is travelling to the right along a stretched rope. The amplitude of the wave is 0.05 m and the oscillator that generates the wave completes 40 vibrations in 20 seconds. A given wave maximum (crest) travels 3.50 m along the rope in 10 seconds. Write the wave equation for vertical displacement in S.I. units.

A) y = 1.00sin(4.4x – 12.6t)
B) y = 0.05sin(35.9x + 12.6t)
C) y = 0.05sin(35.9x – 12.6t)
D) y = 0.05sin(9.0x – 3.14t)
E) y = 0.05sin(9.0x + 3.14t)

A = 0.05 m

40 c in 20 s

T = 0.5 s

$T = \frac{5}{c}$

$= \frac{20}{40}$

$\omega = \frac{2\pi}{T}$

ω = 12.57

$v = \frac{d}{t}$

= 0.35 m/s

$v = \frac{\lambda}{T}$

λ = 0.175 m

$k = \frac{2\pi}{\lambda}$ = 35.9

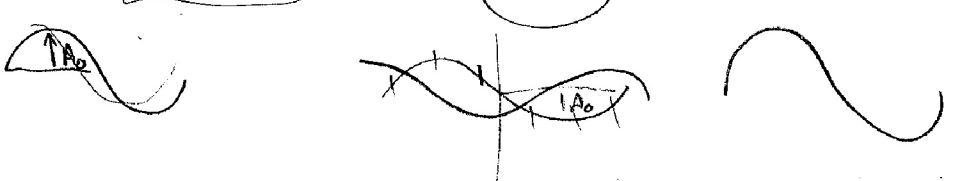
7. [1] Consider two individual, but identical, sinusoidal (harmonic) waves travelling in the same direction at 2.0 m/s. Each wave has an amplitude of A_o and a wavelength of 3.0 m. However, the second wave started 0.5 s after the first wave (from the same origin). When they meet and combine, the amplitude of the resultant wave also equals A_o. Determine the minimum positive value of the phase constant (phase angle).

v = 2.0 m/s

λ = 3.0 m

Δt = 0.5 s

(A) 60° (B) 120° (C) 240° (D) $\frac{\pi}{6}$ rad ~~(E) π rad~~



8. [1] A standing wave is established in a 30 cm length of string that has a mass per unit length of 9.0×10^{-3} kg/m. If the string is subject to a tension of 20 N, determine the oscillator frequency (in Hz) that produces a three-loop pattern on the string, as illustrated.

(A) 236 (B) 2.36 (C) 78.6 (D) 786 ~~(E) 47.1~~

L = 30 cm

μ = 9.0 × 10⁻³ kg

T = 20 N

f = ?

n = 3

$v = \sqrt{\frac{T}{\mu}}$

v =

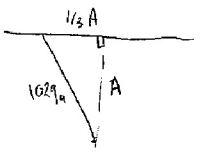
λ = ~~0.1 m~~ ~~1.5 m~~

$f = \frac{n}{2L} v$

v = fλ

[1] An eagle flies horizontally at constant speed and at constant altitude. When the eagle is directly above a shepherd lying on the ground, the shepherd emits a loud whistle of surprise. The bird hears the whistle sound after the bird has travelled a horizontal distance equal to 1/3 its altitude. It took the whistle sound 3.00 s to reach the eagle. If the speed of sound is 343 m/s, find the altitude of the eagle in meters.

(A) 976.2 (B) 925.7 (C) 485.2 (D) 304.7 (E) 30.5

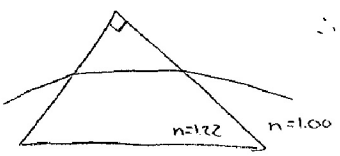


$1029^2 = \sqrt{A^2 + 1/3 A^2}$

10. [1] Green light (λ = 550 nm) strikes a prism which has an apex angle of 90° and an index of refraction of 1.22. The prism is surrounded by air, which has an index of refraction of 1.00. The point of incidence is sufficiently high so that the refracted ray emerges from the opposite sloping side. Assuming that this configuration produces the angle of minimum deviation, determine this angle (of minimum deviation).

(A) 48.0° (B) 7.4° (C) 61.0° (D) 14.7° (E) 29.2°

$\lambda = 550$ nm



$\therefore n \sin(\theta/2) = \sin\left(\frac{\theta + \delta}{2}\right)$

$1.22(\sin 45°) = \sin\left(\frac{90° + \delta}{2}\right)$

11. [1] What is the speed of light (in m/s) in a material for which the critical angle for total internal reflection is 61.0°? The material is surrounded by air, which has an index of refraction of 1.00.

(A) 2.99×10^8 (B) 2.63×10^8 (C) 0.38×10^8 (D) 1.32×10^8 (E) 0.62×10^8

$c = ?$

$\theta_c = 61.0°$

$\therefore \theta_c = \frac{n_c}{n_1}$

$\sin(61°) = \frac{n_2}{1}$

$\therefore n_2$

~~$n = 0.87$~~

$\sin\theta_c = \frac{n_2}{n_1}$

$\sin 61° = \frac{1}{n}$

$n_1 > n_c$

$n = 1.14$

$n = 1.14 \quad \therefore n = \frac{c}{v}$

$v = \frac{c}{1.14} \quad \therefore v = \frac{c}{n}$

$\therefore v = \frac{3.0 \times 10^8}{1.14}$

12. [1] A double-slit experiment has slit spacing 0.12 mm. Determine the slit-to-screen distance (in m) if the bright fringes are to be 5.0 mm apart when the slits are illuminated with 633 nm laser light.

(A) 6.6 (B) 9.5 (C) 0.66 (D) 0.95 (E) 0.15

$d = 0.12$ mm

$L = ?$

$\lambda = 633$ nm

$y = 5.00$ mm

$\frac{y}{L}$

$\sin\theta = \frac{y}{L}$

$\frac{dy}{L} = m\lambda \quad \therefore L = \frac{dy}{\lambda}$